



Paper Id : 1068 - Enhanced Hybrid Deep Learning Model for Cybersecurity Threat Detection Using CNN-BiLSTM and Feature Optimization

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Abstract

Title: *Significance of the Work*

Content:

- Cyberattacks are growing rapidly in number and complexity, targeting modern networks, IoT systems, and critical infrastructure.
- Traditional Intrusion Detection Systems (IDS) fail to detect **zero-day and adaptive threats** due to static, rule-based mechanisms.
- There is a pressing need for **intelligent, real-time, and scalable threat detection models** capable of learning from evolving traffic behavior.
- Deep learning enables automatic feature extraction and pattern recognition in complex network traffic, making it ideal for cybersecurity applications.

Proposed Model & Novelty

Content:

This research proposes an **Enhanced Hybrid Deep Learning Model** combining **PCA**, **CNN**, and **BiLSTM** for intrusion detection.

Novelty:

- Integrates *Principal Component Analysis (PCA)* for optimized feature selection and dimensionality reduction.
- Combines *CNN's spatial learning* with *BiLSTM's temporal sequence modeling*.
- Achieves high detection accuracy with **lower computational cost and fewer parameters** than previous models.

Methodology Steps:

- Data preprocessing and normalization using the **CICIDS2017** benchmark dataset.
- Dimensionality reduction via **PCA**.
- Sequential feature extraction through **CNN–BiLSTM hybrid architecture**.
- Classification using **Softmax layer** for multiple attack types.

Key Results and Conclusions

Content:

- The model achieved **98.2% accuracy, 98.0% F1 score**, and a **1.1% false alarm rate** — outperforming traditional and state-of-the-art hybrid models.
- Visualizations (PCA plots, confusion matrix) confirm strong class separation and generalization ability.
- The model demonstrates **high stability, low false alerts, and suitability for real-time cybersecurity systems** such as IoT networks and smart cities.

Conclusion:

- The PCA–CNN–BiLSTM architecture offers an efficient and scalable framework for modern intrusion detection.
- Future work includes handling encrypted traffic, adversarial resilience, and integration with **edge-based IDS** for distributed protection.

Introduction

◆ Description of the Problem

- The rapid growth of **internet-connected devices**.
- Traditional **Intrusion Detection Systems (IDS)** that rely on fixed rules or signature matching can **only detect known attacks**, failing to identify new or evolving ones such as:
 - Advanced Persistent Threats (APTs)
 - Polymorphic Malware
 - Zero-Day Exploits

◆ Need for a Smarter Detection System

- To defend against these evolving threats, there is a need for **intelligent, adaptive, and real-time intrusion detection mechanisms**.

◆ Proposed Solution

- This work introduces a **hybrid deep learning model** combining:
 - **CNN** for spatial feature extraction.
 - **BiLSTM** for temporal sequence learning.
 - **PCA** for feature reduction and optimization.
- The proposed framework is designed to achieve **real-time performance, lower computational complexity, and high detection accuracy** on modern network datasets.

Related work

Survey of Previous Research:

- Traditional **rule-based and signature-based IDS** effectively detected known threats but failed against **zero-day and adaptive attacks**.
- Machine Learning models (e.g., **Random Forest, SVM**) improved detection but required **manual feature engineering**.
- Deep Learning models like **CNN–GRU, CNN–RNN**, and **LSTM–SVM** achieved better accuracy on datasets such as **CICIDS2017** and **UNSW-NB15**.
- *Abdulkareem et al. (2023)* and *Hou et al. (2019)* proposed hybrid DL models with strong performance in IoT environments.

Comparison of Methods and Techniques:

- **CNN-based** models learn spatial dependencies well but ignore sequential attack patterns.
- **LSTM/GRU** capture temporal features but are prone to **overfitting** and **high training time**.
- **Metaheuristic-optimized hybrids** improved accuracy but were **computationally expensive** and unsuitable for **real-time detection**.
- Hence, an optimized **CNN–BiLSTM** hybrid is needed to balance **accuracy and efficiency**.

Related work

Key Contributions and Limitations of Prior Works:

- Deep hybrid models improved accuracy and false alarm rates.
- However, most were **complex, resource-intensive**, and lacked **scalability**.
- Ensemble and attention-based networks enhanced feature learning but **increased architectural depth and computation time**.

Gaps in Existing Research:

- Insufficient focus on **lightweight, real-time IDS** with **low false positive rates**.
- Limited use of **feature optimization techniques** like PCA in deep architectures.
- Few models integrate **both spatial and temporal learning** efficiently.

Methodology

Raw Network Data Collection:

Dataset used: **CICIDS2017**, which includes both **benign** and **malicious** traffic generated in realistic network environments.

Covers multiple attack categories: DoS, DDoS, Port Scan, Brute Force, Web Attacks, and Infiltration.

Data Cleaning:

Removed incomplete records containing **NaN**, **Inf**, or **missing values**.

Dropped **irrelevant fields** such as IP addresses and timestamps that don't contribute to attack detection.

Label Encoding:

Converted **categorical attack names** (e.g., "DoS Hulk," "PortScan") into **numeric labels** for model training.

This step ensures the data can be processed by neural network layers.

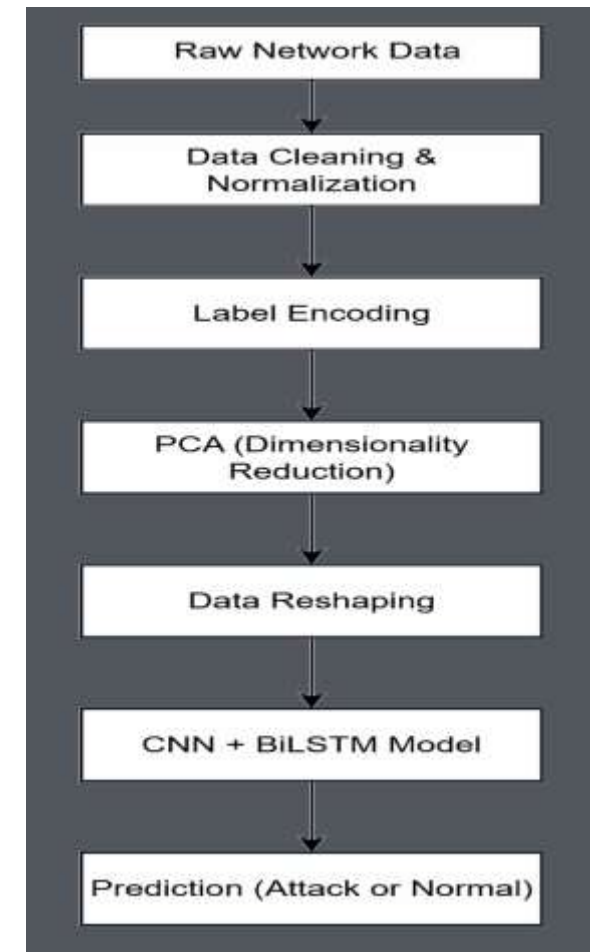
Feature Normalization:

Applied **z-score normalization** to bring all numerical features to a common scale (mean = 0, standard deviation = 1).

This prevents large-valued features from dominating the learning process and ensures stable convergence.

Final Preprocessed Dataset:

After these steps, the dataset becomes **clean, consistent, and numerically encoded**, ready for **PCA-based feature optimization**.



Proposed Hybrid Model Architecture – CNN + BiLSTM + Softmax

Input Layer: Receives preprocessed and PCA-reduced feature data.

1. CNN Layers:

- Apply convolution filters to extract **spatial feature patterns** from input data.
- Identify localized traffic behaviors like packet bursts, protocol anomalies, or header irregularities.

2. Pooling & Dropout Layers:

- Reduce dimensionality and prevent overfitting by randomly deactivating neurons.

3. BiLSTM Layers:

- Process sequential data **in both forward and backward directions** to understand time-based relationships.
- Capture temporal dependencies between network events (e.g., request-response sequences).

4. Fully Connected (Dense) Layer:

- Integrates learned spatial-temporal features and prepares for final classification.

5. Softmax Output Layer:

- Produces probabilistic predictions across multiple classes (e.g., *Normal, DoS, DDoS, PortScan, Brute Force*).

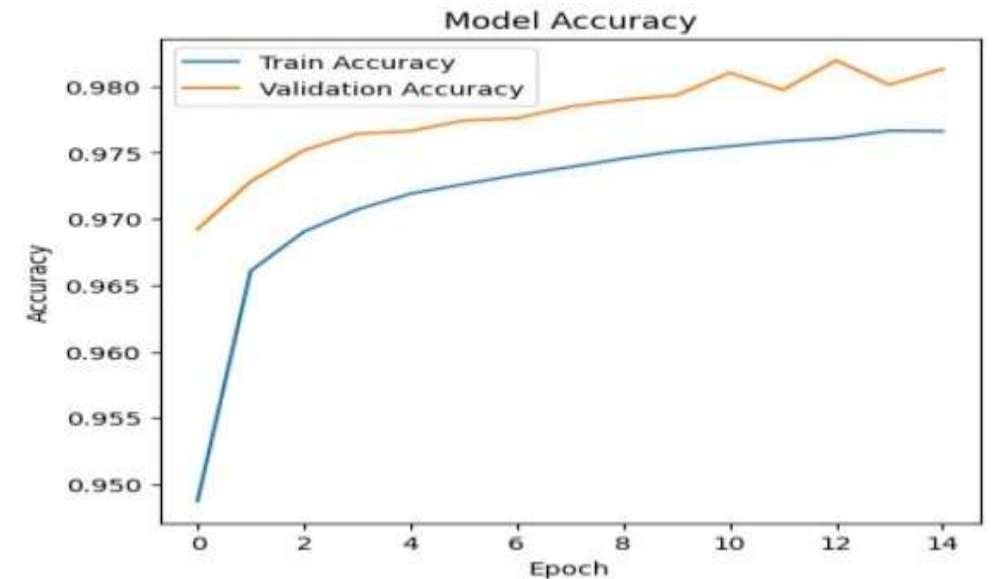
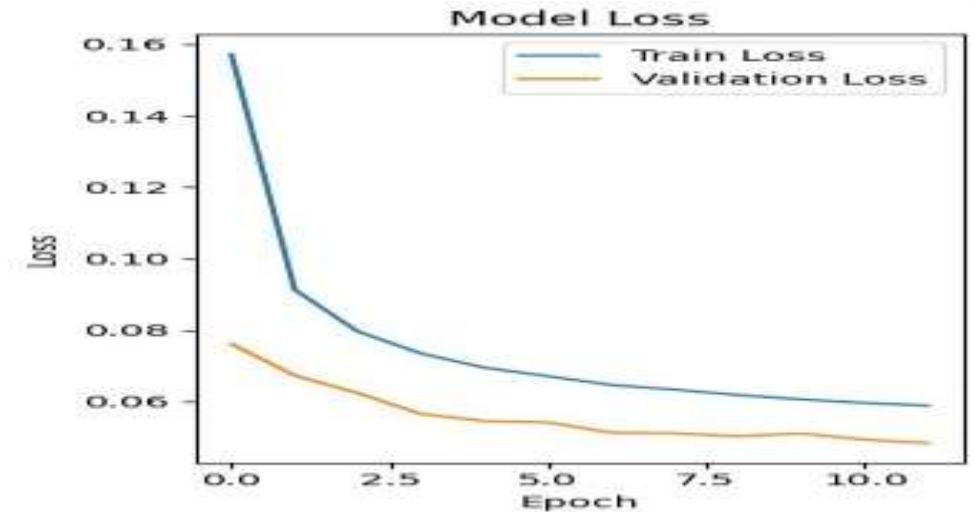
Results/discussion

The proposed **PCA–CNN–BiLSTM hybrid model** successfully overcomes the limitations of traditional Intrusion Detection Systems (IDS) by providing:

- **High accuracy** in detecting both known and unknown attacks.
- **Reduced false alarm rate**, ensuring reliable real-time performance.
- **Lower computational complexity** through PCA-based feature optimization.

Major Advances:

- Effective combination of **spatial (CNN)** and **temporal (BiLSTM)** learning for comprehensive traffic analysis.
- **Faster convergence** and **better generalization** achieved via feature reduction with PCA.
- **Superior accuracy and stability** compared to previous deep learning models tested on the same dataset.



Conclusion

Conclusion:

- This research successfully developed an **enhanced hybrid intrusion detection model** combining **PCA, CNN, and BiLSTM** to achieve **efficient, accurate, and real-time cyber threat detection**.
- The model attained **98.2% accuracy** and a **1.1% false alarm rate**, demonstrating significant improvement over traditional and existing deep learning approaches.
- The integration of **PCA-based feature optimization** minimized redundancy, while **CNN–BiLSTM** captured both **spatial and temporal features** of network traffic.

Benefits:

- **High performance:** Accurate classification of multiple attack types with minimal false alarms.
- **Efficiency:** PCA reduces feature dimensions, resulting in faster training and lower computation cost.
- **Scalability:** Model adaptable to large and complex network environments such as IoT and smart grids.

Future Scope

Shortcomings:

- Model accuracy may drop when dealing with **encrypted or highly obfuscated traffic**.
- **Retraining required** when new attack patterns or network behaviors emerge.
- Computational demand still higher than traditional ML models for real-time edge devices.

Future Scope:

- Extend the framework to handle **encrypted and adversarial network traffic**.
- Integrate with **edge or fog computing environments** for distributed intrusion detection.
- Incorporate **attention mechanisms** or **transformer-based architectures** for better contextual learning.
- Develop a **real-time deployment prototype** for industrial or smart city cybersecurity systems.

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